Summary of Topics

1. Scheduling
    - Know how to construct a digraph from a dependency list
    - Know the meaning of the following
        (a) Project digraph
        (b) schedule
        (c) priority list
        (d) processor
        (e) critical path
        (f) critical time
        (g) idle time
        (h) decreasing time algorithm/priority list
        (i) backflow algorithm
        (j) critical time priority list
    - Be able to create a schedule from a digraph or table of dependencies and a priority list, and determine the finishing time, idle time, and whether the schedule is optimal
2. Cryptography
    - Know how to encrypt and decrypt a message using
        - a Caesar cipher
        - an alpha-numeric cipher given a substitution mapping
        - a transposition cipher given the length of a row
        - a transposition cipher given an encryption keyword
        - a shifting substitution / progressive Caesar cipher
        - a Vigenère cipher
        - a double transposition cipher (given two keywords)
    - Know how to use letter frequency to decrypt a message without an encryption key.
    - Know which encryption methods do or do not preserve letter frequency.
    - Be able to articulate strengths and weakness in each of the encryption methods learned.
    - Be able to recognize and explain why some keywords are good and some are not.

Tabula Recta

A Tabula Recta is provided on the last page. It will be provided on the exam as well.

Scheduling Sample Problems

1. The table below contains the tasks to be completed for a project.

task	time (in minutes)	dependencies/ prerequisites
A	2	
B	9	
C	3	
D	7	A
E	1	C
F	6	A
G	4	E
H	8	D,F
I	5	G
J	1	B,H,I

   (a) Next to the table, create a **digraph** representing the project.

   (b) Find a **critical path** and its corresponding **critical time**. Explain the significance of the critical time.

   (c) Create a priority list using the **decreasing time** algorithm. (That is, create a priority list ordered by decreasing time.)

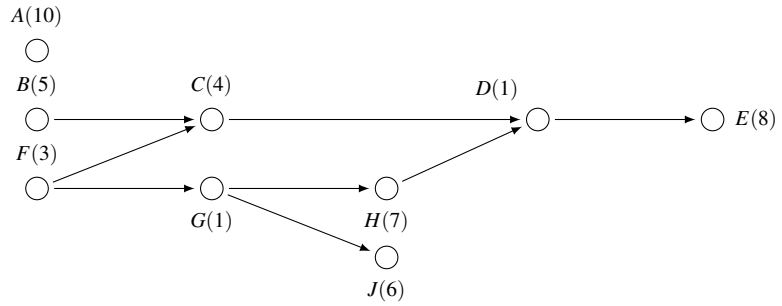
   (d) Use the **Backflow algorithm** to assign critical numbers to each vertex in the graph above.

   (e) Create a priority list using the critical time algorithm. (That is, create a priority list ordered by decreasing **critical** time.)

   (f) How long would it take for one processor to complete all the tasks? _____

   (g) Is it possible for 2 processors to complete a schedule in exactly the critical time? Why or why not? Provide a supporting calculation.

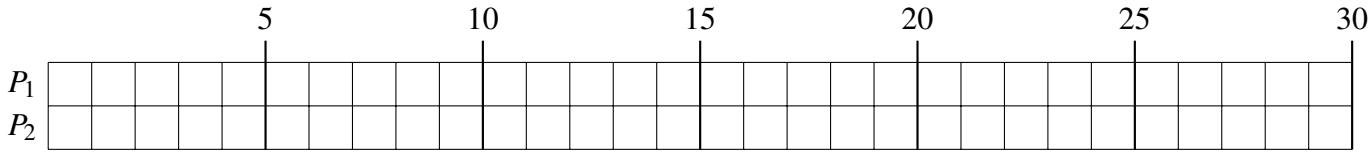
2. Use the diagraph below and the given priority list to construct a schedule using two processors.



Construct a schedule using the priority list

$$F,\ B,\ G,\ C,\ A,\ H,\ J,\ D,\ E$$

time	
ready	
done	



Finishing time: _____

Idle time: _____

Critical path / time: _____

Is your schedule optimal? How do you know?

Cryptography Sample Problems

1. Encrypt the message GO TO BERLIN using

   (a) a Caesar cipher with a shift of 24 (A $\rightarrow$ Y)

   (b) a transposition cipher given a row length of 5.

   (c) a transposition cipher given encryption keyword `STOP`.

   (d) a shifting substitution cipher (progressive Caesar cipher) starting with a shift of 6 (A to G).

(e) a Vigenére cipher with key word `STOP`.

(f) a double transposition cipher with first key word `BLUE` and second key word `GOLD`.

2. Using the alpha-numeric substitution below to answer the questions.

original	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
maps to	9	8	7	6	5	4	3	2	1	0	D	C	B	A	H	G	F	E

original	S	T	U	V	W	X	Y	Z	0	1	2	3	4	5	6	7	8	9
maps to	L	K	J	I	P	O	N	M	T	S	Q	R	V	U	Y	X	W	Z

(a) Encrypt the plaintext: RIGHT 10 LEFT 25

(b) Decrypt the ciphertext: K5OKT TQXW

3. Decrypt each encrypted text using the specified mechanism.

   (a) ciphertext: NJCYQC
   encryption mechanism: a Caesar cipher with a shift of 24 (A to Y)

   (b) ciphertext: DFTOO ANRBO GCTED
   encryption mechanism: a transposition cipher given a row length of 5

   (c) ciphertext: DMFAT 3RYME IIIRS D
   encryption mechanism: a transposition cipher given encryption keyword STOP

   (d) ciphertext: HYQWQ
   encryption mechanism: a shifting substitution cipher / progressive Caesar cipher starting with a shift of 6 (A to G)

(e) cipher text: FHHTK ASTL
encryption mechanism: a Vigenère cipher with key word `STOP`

(f) cipher text: OEAOM NTSLD
encryption mechanism: a double transposition cipher with first key word `BLUE` and second key word `GOLD`.

4. Which of the encryption methods in problems 1 and 2 above preserve letter frequency?

		0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
		A	**B**	**C**	**D**	**E**	**F**	**G**	**H**	**I**	**J**	**K**	**L**	**M**	**N**	**O**	**P**	**Q**	**R**	**S**	**T**	**U**	**V**	**W**	**X**	**Y**	**Z**
0	**A**	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	**B**	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
2	**C**	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
3	**D**	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
4	**E**	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
5	**F**	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
6	**G**	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
7	**H**	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
8	**I**	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
9	**J**	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
10	**K**	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
11	**L**	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
12	**M**	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
13	**N**	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
14	**O**	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
15	**P**	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
16	**Q**	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
17	**R**	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
18	**S**	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
19	**T**	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
20	**U**	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
21	**V**	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
22	**W**	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
23	**X**	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
24	**Y**	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
25	**Z**	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y